1 POWER INPUT, 5 V AND 3.000 V REFERENCE

9 V / ≈+1.5 A desktop adapter (5.5/2.1 mm, centre +).
Q1 = reverse-polarity protection, F1 = resettable fuse, D2 = TVS.

J1 DC IN
F1 2A PTC
Q1 SI2302
D2 SMAJ5.0A
U1 TBH05A80T
C1 220uF 25V
C2 220uF 25V
C3 10uF
C4 10uF
C5 100nF
C6 1uF
C7 10uF
C8 100nF
C9 1uF
C10 1uF
U2 REF3030
PWR_FLAG
PWR_FLAG

2 SETPOINT, SOFT START AND TTL GATING (scale 4 V/A)

VFS = 2.40 V = 600 mA full scale (2P1 closed); 2.80 V = 700 mA).
C11 = 910 (91) gives an 80 ms soft start; Q3 discharges it when disabled.
U5 selects PULSE (TTL high) or IDLE (TTL low, 5.10k mA).

R10 10k 0.1%
R11 40.2k 0.1%
JP1 LMAX 700mA
R12 4.02k
C11 10uF
U3 TLV9062
U3B TLV9062
R13 100
C12 220pF C0G
R14 20k
R15 100k
C13 100nF
U4A TLV9062
R16 1k
R17 1k
C14 100nF
C15 100nF C0G
R18 100k
C16 100nF
U5 74LVC1G157
Q3 2N7002
C17 100nF
C18 100nF
C19 1uF
R30 330
C31 220pF C0G
R37 330k
R38 47k
Q2 IRF9022V4
R42 8.2
C20 100nF
D3 SS 83950

3 TTL INPUT (BNC, 10 k to GND, 3.3/5 V logic)

Schmitt trigger with TTL thresholds; unconnected input = IDLE.

J2 TTL IN
D4 BAT54S
U13 74AHCT1G14
R22 10k
R21 220
C15 100pF C0G
C17 100nF

4 HIGH-SIDE CURRENT SOURCE (laser cathode at GND)

LLD = VSP × (R32/R31) // Rhd = VSP / ≈ V/A.
U7A error amp senses Rhd directly (Kelvin via NTL/NCD); Rh1 degenerates Q2.
Q6 clamps Q2 off unless EN_LOOP is high (fail-safe; QT/QR release 1Ω.
Q5 forces U7A high while disabled so the integrator cannot wind up.

U7A OPA2192
U7B OPA2192
R31 3k 0.1%
R32 150 0.1%
C18 100nF
C30 100pF C0G
R35 10k
R36 100k
Q5 2N7002
Q6 2N7002
R39 100k
R48 47k
R49 1.0
R50 47k
R51 1.0
Q4 NDS9401A
Q7 NDS9401A
R71 RefTie
R72 RefTie
C19 1uF
R33 1k
C31 220pF C0G
R37 330k
R38 47k
Q8 2N7002
R42 8.2
C20 100nF
U8 INA180A1
D3 SS 83950
Q2 IRF9022V4
R34 330
R38 47k
R42 8.2
D3 SS 83950
Q4 NDS9401A
Q7 NDS9401A
R71 RefTie
R72 RefTie
R49 1.0
R51 1.0

5 OUTPUT, SHORTING RELAY AND CROWBAR

K1 NC contacts short the laser whenever the driver is not armed or unpowered.
Q10 crowbar shunts I_OUT within 1 µs of a fault; D6/D7 reverse and surge clamp.
D99 follows Thorlabs LD-driver 'CC' pinout: 8 = anode, 3 = cathode/GND, 1–5 interlocks.

K1 660 660 5 3 4 2 1
K1 5VDC-2F-Y-5V
Q9 2N7002
D6 SS34
D5 1N5148WS
Q10 NDS9410A
R43 100k
R44 22
R42 100
C32 47nF
D99 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
D99 DB15-HD-M
D7 SMAJ22A

6 CURRENT AND SETPOINT MONITORS (1 V/A into high-Z)

I_MON from INA180 (≈ V/A) // ≈ SET MON shows the PULSE setpoint // ≈.

U6A TLV9062
U10B TLV9062
U10A TLV9062
R8 3k 0.1%
R45 3k 0.1%
C21 100nF
R9 14 0.1%
R46 14 0.1%
R47 49.9
R20 48.9
J7 SET MON
J8 I_MON

7 PROTECTION COMPARATORS

A over-current = 800 mA. B: compliance / open laser (gate = 2 V for > 10 us).
C: Q2 heatsink > 85 C (NTC RT1). D: enclosure > 19 mA (gets the EMISSION LED for ~0.15 s.

U11A LMV901
U11B LMV901
U11C LMV901
U11D LMV901
U11E LMV901
R61 18k
R54 100k
R55 11k
R58 100k
R59 100k
R60 1.5k
R61 1k
C33 100pF C0G
C34 100nF
R56 10k
C35 100nF
C23 100nF
C24 100nF
R62 1k
D11 SML-D12V8W
R50 330
R52 300k
R53 100k

8 INTERLOCKS, FAULT LATCH AND ENABLE SEQUENCING

Any fault, open interlock or power-up sets the FAULT latch; reset by switching ENABLE off.
EN_REQ arms relay K1 at once; EN_LOOP releases the current source ~17 ms later.

SW1 ENABLE
R70 10k
C42 10nF
C40 100nF
J6 INTERLOCK
R13 4.7k
C41 100nF
R77 3.3k
U12A 74HC14
U12B 74HC14
U12C 74HC14
U12D 74HC14
U12E 74HC14
U12F 74HC14
D10 BAT54A
R73 10k
C44 1nF
R78 100k
C43 2.2nF
C25 100nF
D14 1N5148WS
U13A 74HC132
U13B 74HC132
U13C 74HC132
U13D 74HC132
D13 1N5148WS
R79 100k
C45 220nF
D15 1N5148WS

9 FRONT-PANEL INDICATORS

D3 POWER green
D16 ARMED yellow
R7 2.2k
R82 1k
D12 FAULT red
D8 EMISSION blue
FB1 1k

CLASS 4 LASER — read README safety section before use
Laser: DTR BDR-209 405 nm (500 mA nominal, 600 mA max), CATHODE AT GND
Generated from tools/design.py by tools/gen_sch.py
ion-trap lab photoionisation 405 nm driver
Sheet: /
File: ld405-driver.kicad_sch
Title: 405 nm laser diode pulse driver (BDR-209)
Size: A1 | Date: 2026-10-28 | **Rev: 1.0**
KiCad 10.0.1 | ID: 1/1

10 IC POWER PINS

U1C TLV9062
U4C TLV9062
U6B SN74LVC1G3157
U7C OPA2192
U10C TLV9062
U11E LMV901
U13E 74HC14
U13C 74HC132

11 MOUNTING HOLES

H1 M3
H2 M3
H3 M3
H4 M3